

Data Security

Master's in Cybersecurity

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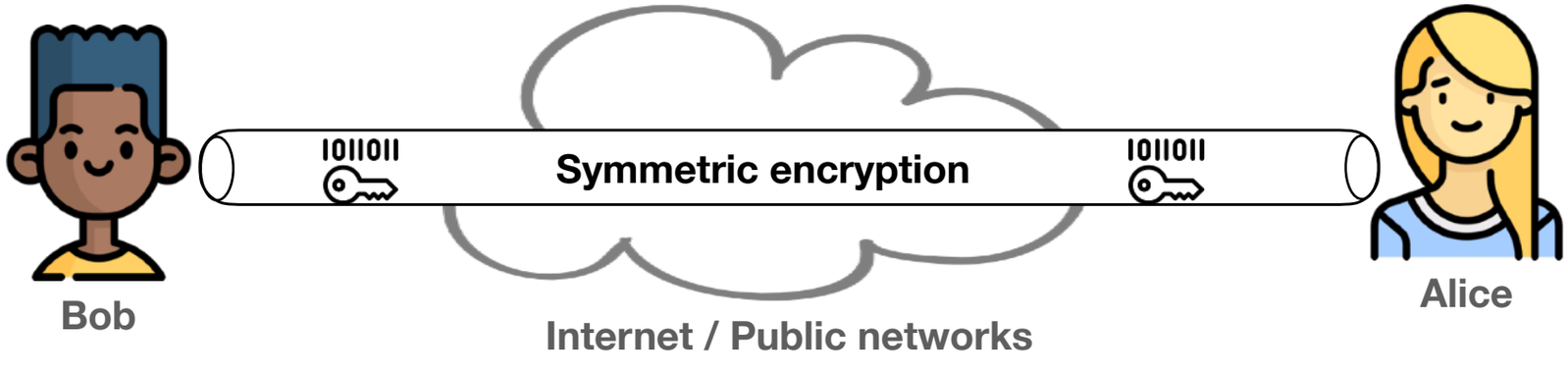
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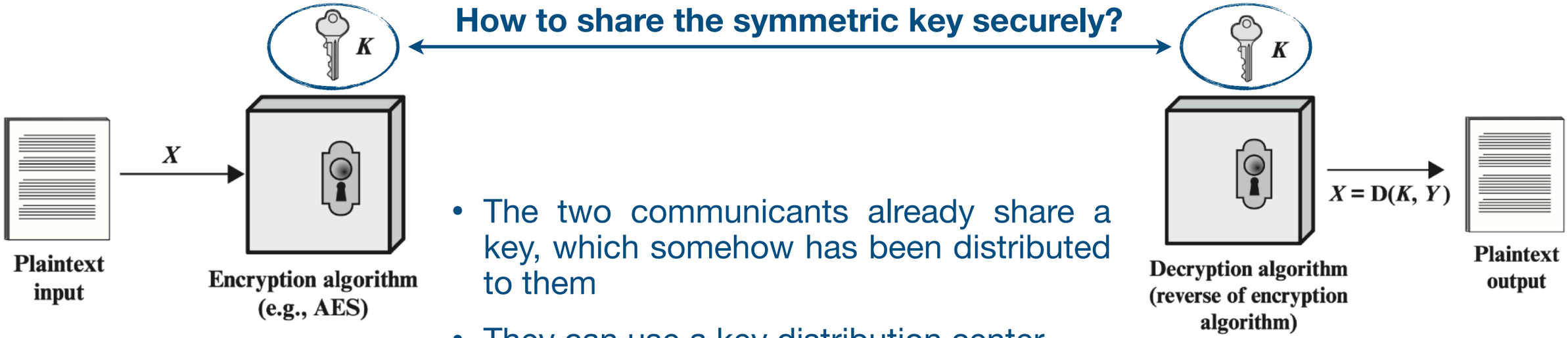
Data Security

Objectives

- Understand the properties of public-key cryptography
- Identify its main applications and limitations
- Understand key exchange schemes based on public-key cryptography
- Understand how the RSA algorithm works

Recap





- The two communicants already share a key, which somehow has been distributed to them
- They can use a key distribution center

  → A third party has access to the key that should be known only by the two communicants.

Asymmetric Cryptography

Concepts

Cryptosystems in which enciphering and deciphering are governed by distinct keys, `E` and `D`, such that computing `D` from `E` is computationally infeasible

Since the enciphering key E can be publicly disclosed without compromising the deciphering key D, such systems are also known as **public-key cryptography.**

They are widely used in…

- Key exchange protocols
- Digital signature
- Authentication

Asymmetric Cryptography

Principles

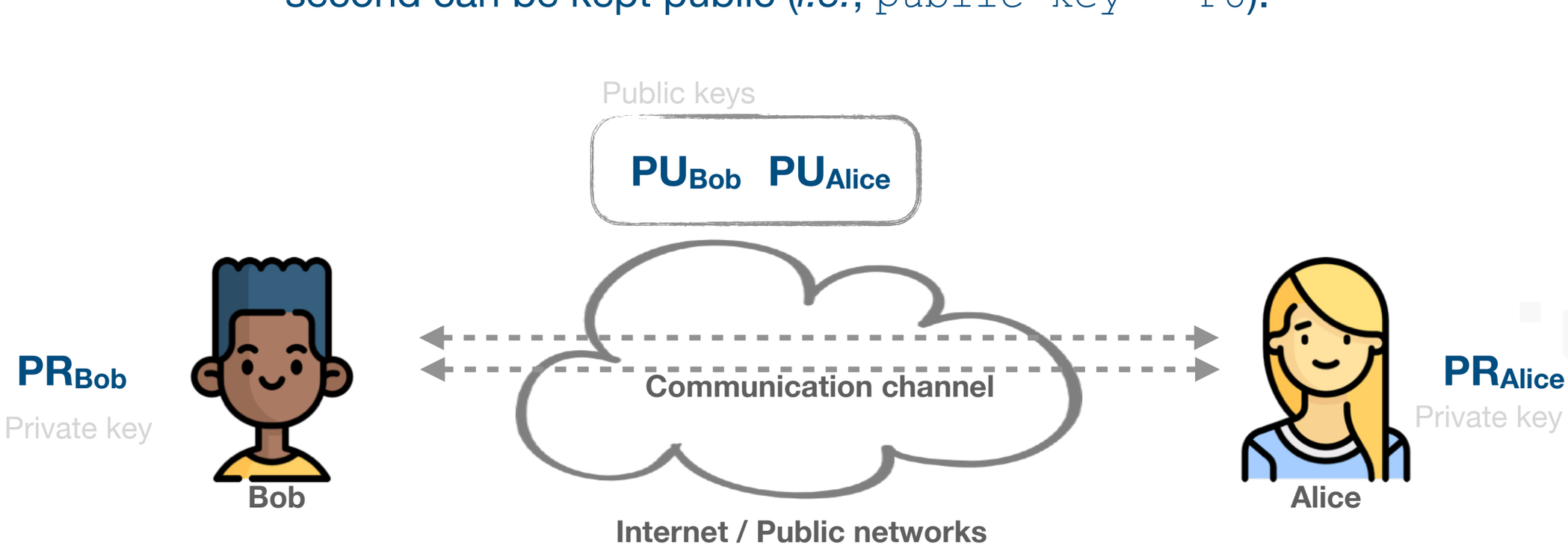
- Asymmetric algorithms rely on one key for encryption and a different but related key for decryption. They have two main characteristics:
  - It is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and the encryption key;
  - Either of the two related keys can be used for encryption, with the other used for decryption.
- A tighter specification is used for *one-way functions with secret*: **`Trapdoor Permutation:`**
  - Injective function that has an efficient calculation algorithm;
  - Whose the inverse calculation is an intractable problem;
  - But anyone with additional information (*i.e.*, the secret) can calculate its inverse.

Asymmetric Cryptography

Principles

By using different keys for the encryption and decryption operation, it is possible to overcome the problem of pre-distribution of keys.

One key must be kept secret (*i.e.*, `private key - PR`), while the second can be kept public (*i.e.*, `public key - PU`).



Asymmetric Cryptography

Terminology

Asymmetric cryptography: A cryptosystem that uses separate keys, one to encrypt or digitally sign data and one for decrypting the data or verifying the digital signature.

Asymmetric keys: Two related keys, a public and a private key, that are used to perform complementary operations, e.g., encryption and decryption or signature and signature verification.

Public key certificate: A digital document issued and digitally signed by the private key of a Certification Authority that binds the name of a subscriber to a public key.

Public key Infrastructure (PKI): A set of policies, processes, server platforms, and software used to issue, maintain, and revoke certificates and public-private key pairs.

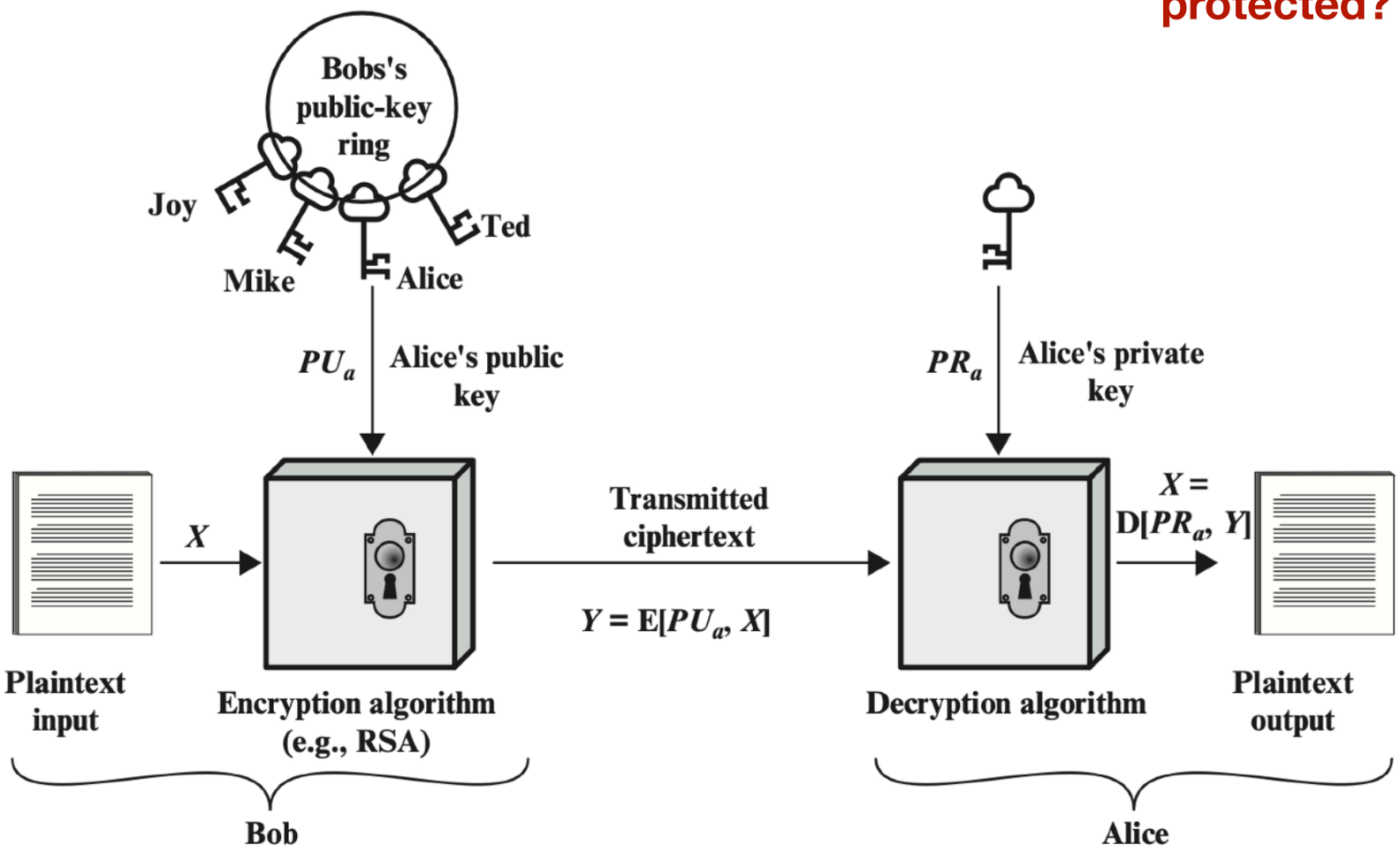
→ Covered in class 09

Asymmetric Cryptography

Concepts

Encryption with public key

Which security properties are protected?



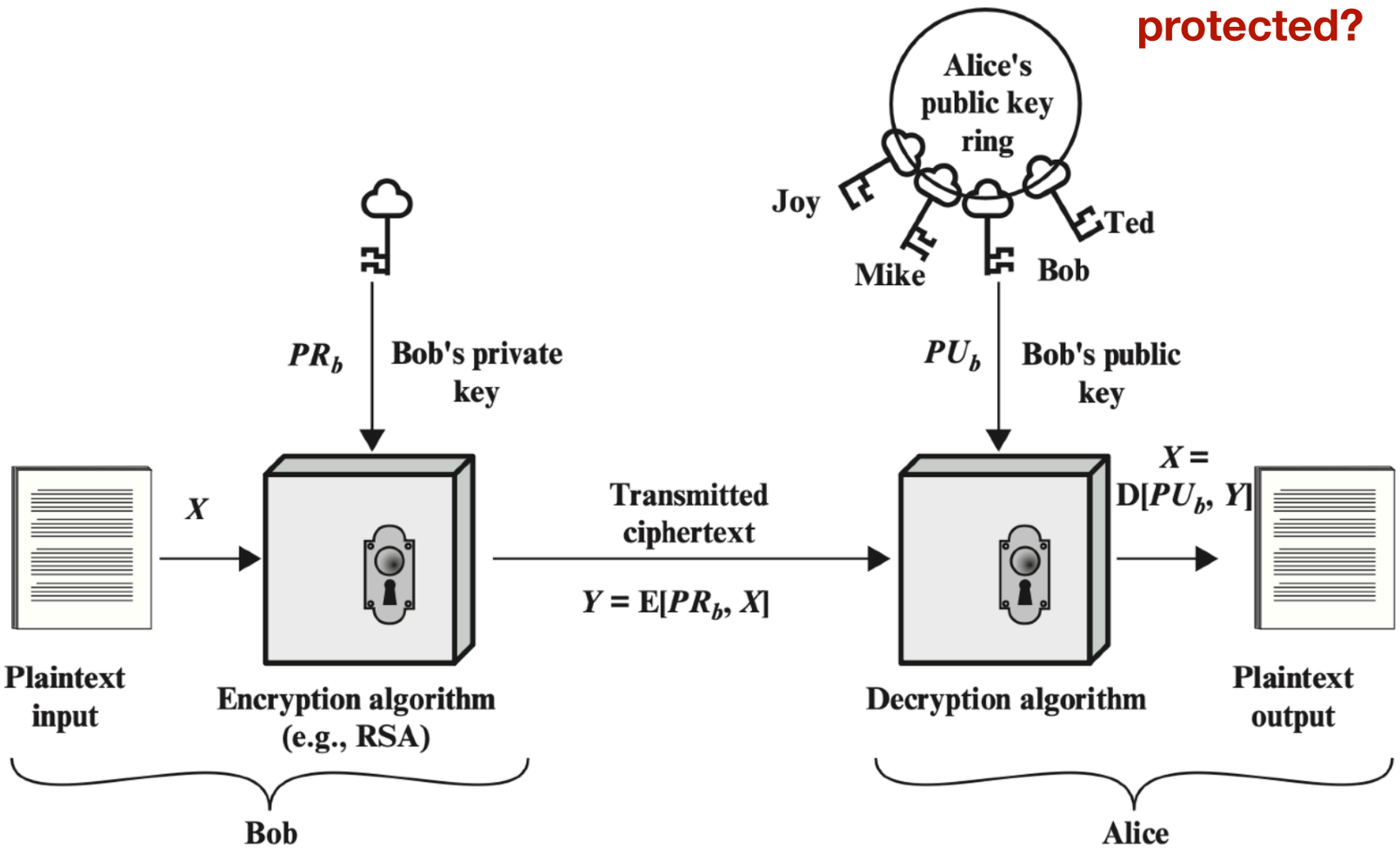
Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Asymmetric Cryptography

Concepts

Encryption with private key

Which security properties are protected?



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Asymmetric Cryptography

Concepts

- Although better than symmetric cryptosystems, public-key cryptography requires protocols for key distribution, generally involving a central agent.

- One common misconception is that public-key encryption is more secure from cryptanalysis than symmetric encryption.

- Another misconception is that asymmetric encryption made symmetric encryption obsolete.

  - For the same level of security, asymmetric encryption is several orders of magnitude less efficient than symmetric encryption.

  - For this reason, they are typically used in conjunction and not alternatively.

- Public-key algorithms are based on mathematical functions rather than on substitution and permutation.

Asymmetric Cryptography

Requirements

- It is computationally easy for a party **B** to generate a key pair ($\mathbf{PU_b}$, $\mathbf{PR_b}$)

- It is computationally easy for a sender **A**, knowing the public key and the message to be encrypted, **M**, to generate the corresponding ciphertext.

$$\mathbf{C = E(PU_b, M)}$$

- It is computationally easy for the receiver **B** to decrypt the resulting ciphertext using the private key to recover the original message.

$$\mathbf{M = D(PR_b, C) = D[PR_b, E(PU_b, M)]}$$

- It is computationally infeasible for an adversary, knowing the public key, $\mathbf{PU_b}$, to determine the private key, $\mathbf{PR_b}$.

- It is computationally infeasible for an adversary, knowing the public key, $\mathbf{PU_b}$, and a ciphertext, **C**, to recover the original message, **M**.

- The two keys can be applied in either order.

Asymmetric Cryptography

Applications

The first practical application of public key algorithms was key exchange protocols, particularly the Diffie-Hellman scheme

Diffie-Hellman Key Exchange

The purpose of the algorithm is to allow communication parties, who have no prior knowledge of each other, to exchange shared secret keys over an insecure channel.

- Its effectiveness relies on the difficulty of computing discrete logarithms.

  Briefly, given a, b and a prime number n, find x such that
  $a^x \bmod n = b$

- The exponent x is referred to as the *discrete logarithm* of b for the base a, $\bmod n$.

The algorithm is limited to exchanging secret values used to derive the symmetric secret key.

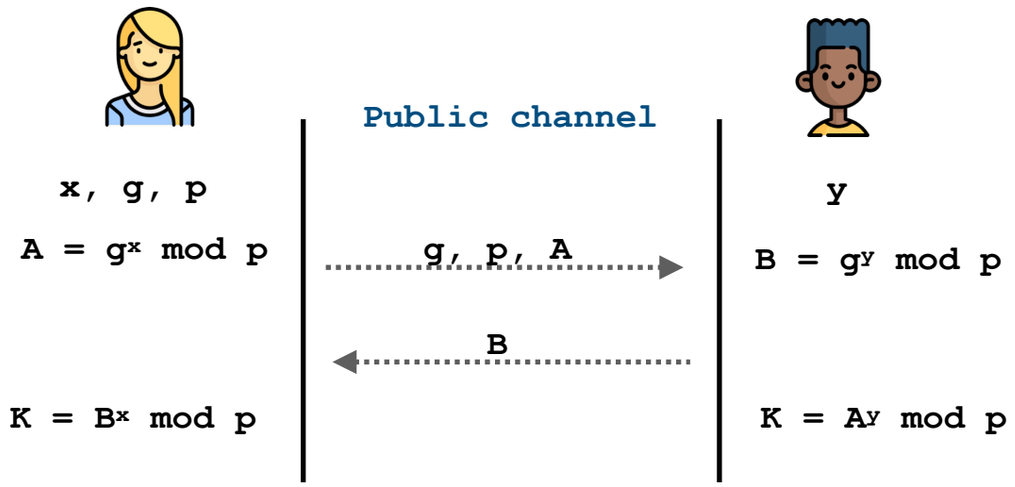
Asymmetric Cryptography

Applications

Diffie-Hellman Key Exchange

Simplified scheme

Parameters: A prime number p and a generator g (that is a *prime root* of p)



p is typically a big prime number, e.g., a 2048-bit number

g can be a small prime number, e.g., 2 or 3

Description

- Alice generates an integer $1<x<p$ and sends $\mathbf{g^x \bmod p}$ to Bob
- Bob generates an integer $1<y<p$ and sends $\mathbf{g^y \bmod p}$ to Alice
- The shared secret: $\mathbf{(g^y \bmod p)^x \bmod p = (g^x \bmod p)^y \bmod p}$
  - In practice $\mathbf{K = K' = g^{xy} \bmod p}$

Asymmetric Cryptography

Elliptic-Curve Cryptography (ECC)

- Typically used for key agreement, digital signatures, and pseudo-random generators, it is based on the *discrete logarithm* problem and can be expressed in any *finite field*

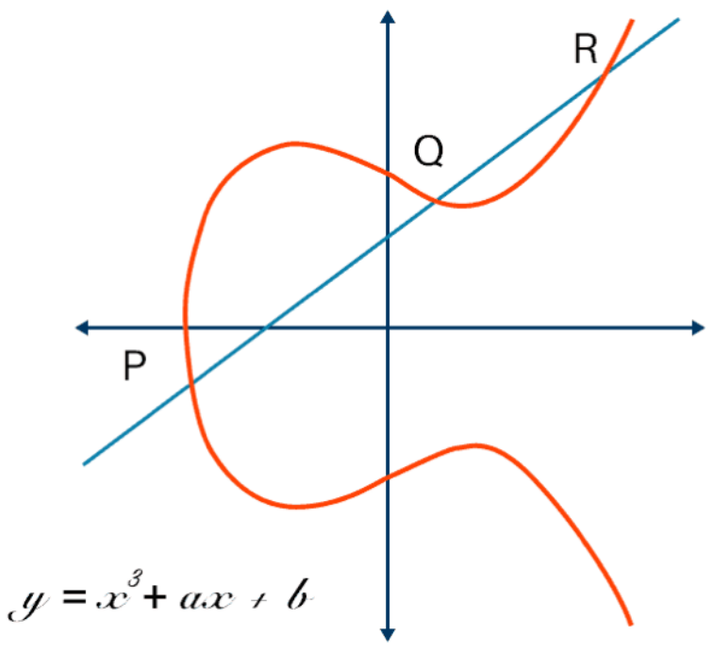
  `GF(p)` <- Prime field

  `GF(p^n)` <- Binary field

- In particular, it is possible to express exponentiation in the cyclic group defined by an elliptic curve on the considered field
- Offers equivalent security with shorter key lengths
  - Efficient for constrained devices and bandwidth

Symmetric Key Algorithms	Diffie–Hellman, Digital Signature Algorithm	RSA (size of *n* in bits)	ECC (modulus size in bits)
80	$L = 1024$ $N = 160$	1024	160–223
112	$L = 2048$ $N = 224$	2048	224–255
128	$L = 3072$ $N = 256$	3072	256–383
192	$L = 7680$ $N = 384$	7680	384–511
256	$L = 15{,}360$ $N = 512$	15,360	512+

Note: L = size of public key, N = size of private key.

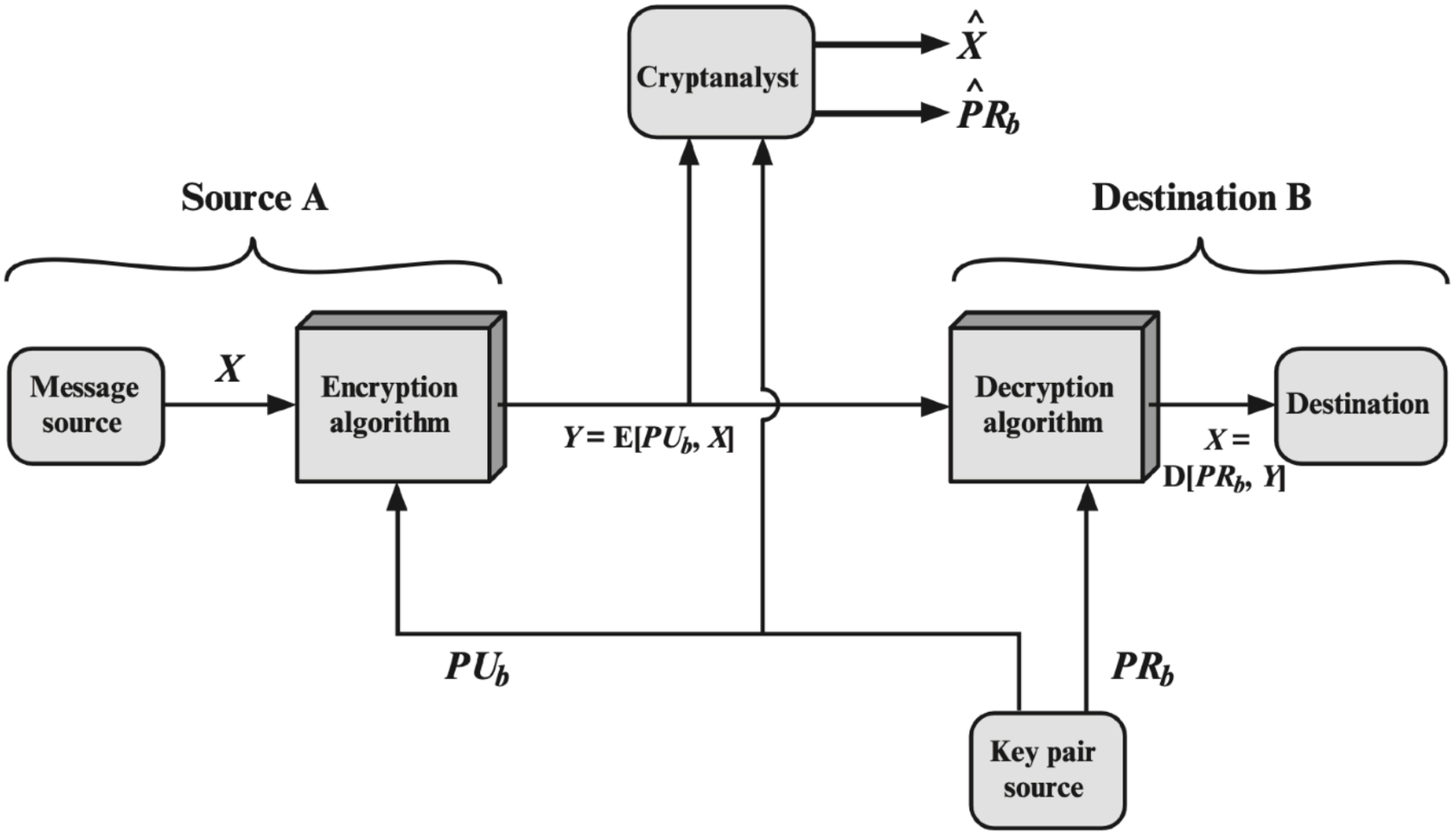


Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Asymmetric Cryptography

Applications

Public-Key Cryptosystem: Confidentiality

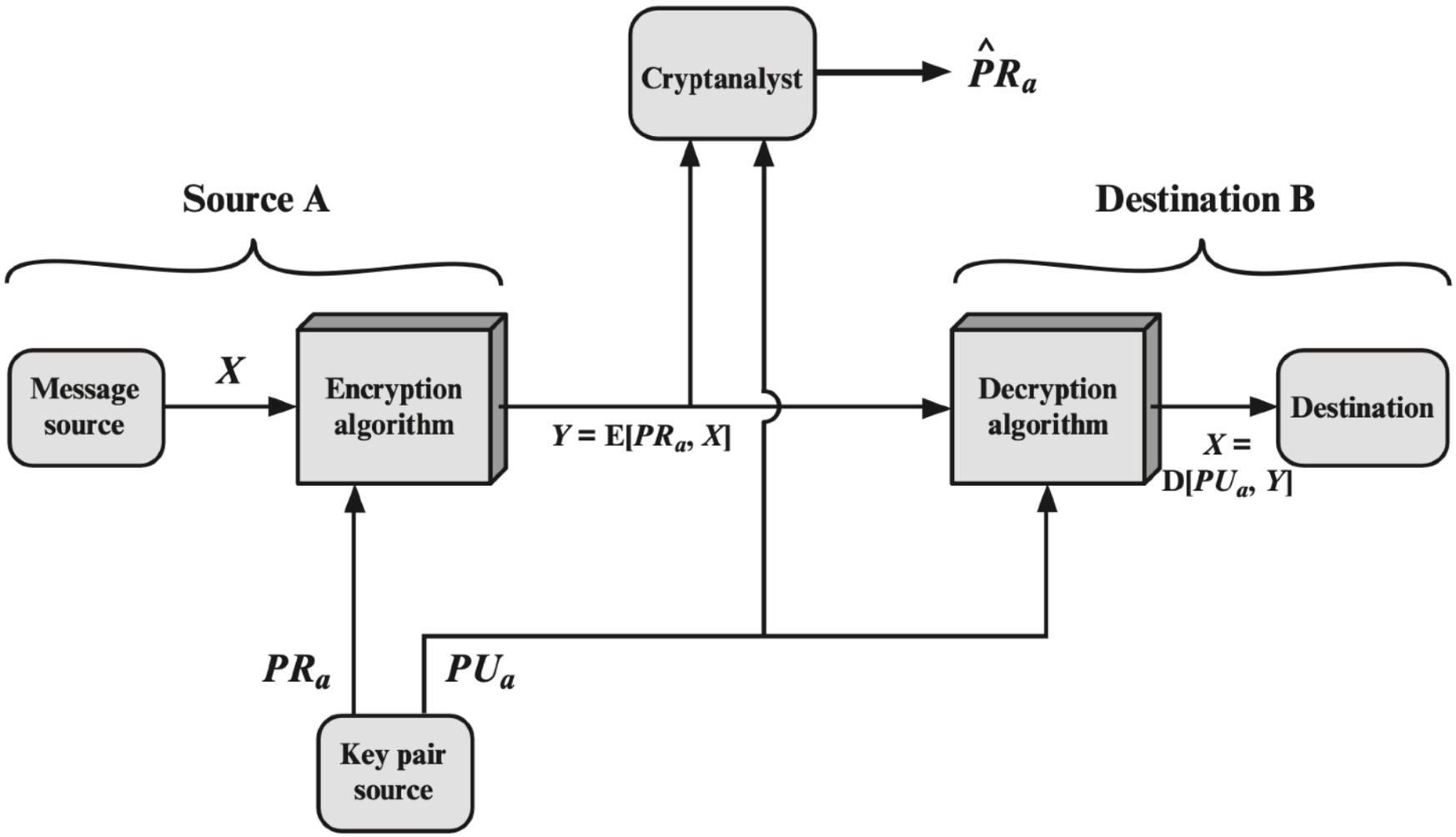


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Asymmetric Cryptography

Applications

Public-Key Cryptosystem: Authentication

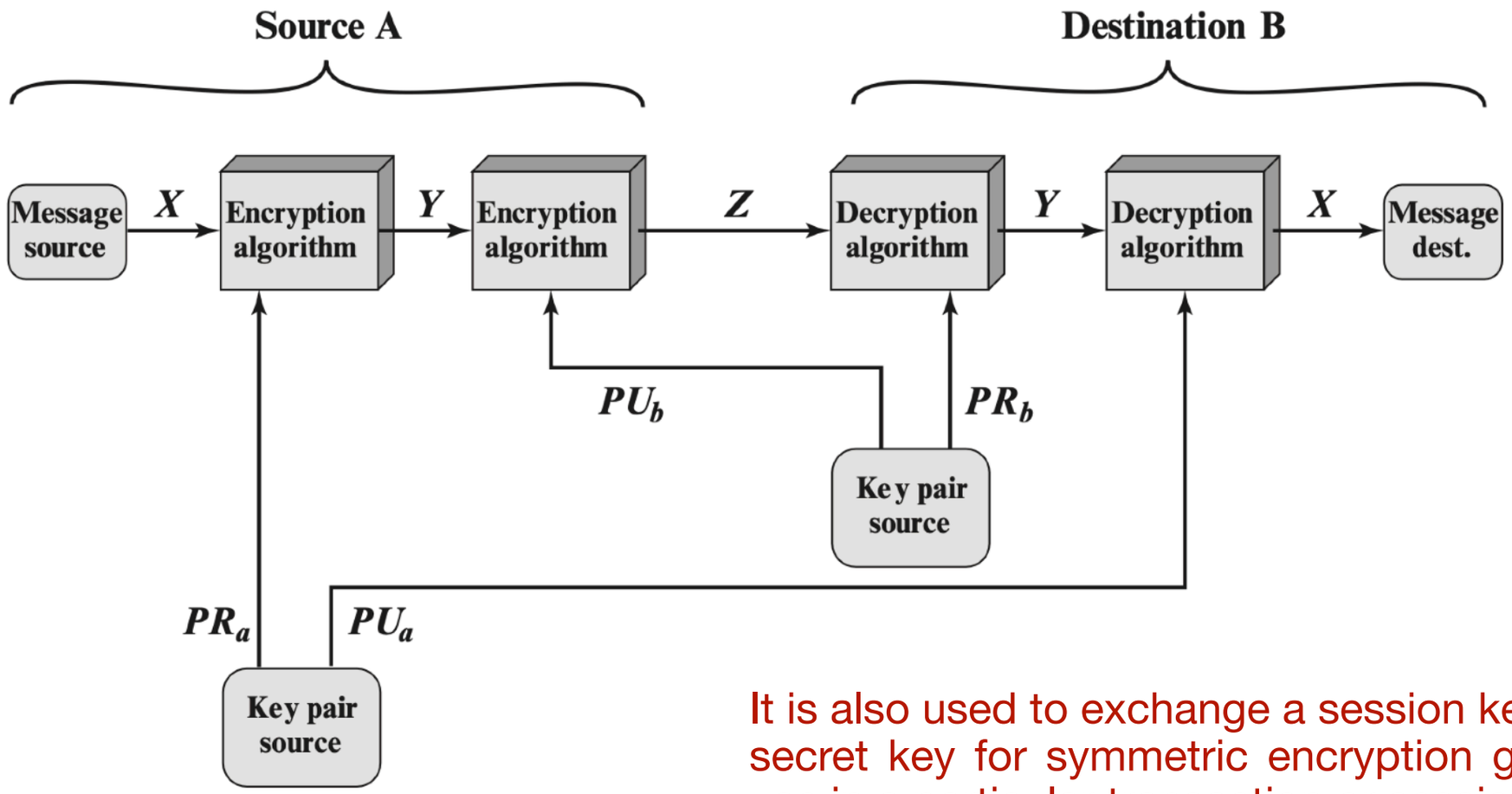


Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Asymmetric Cryptography

Applications

Public-Key Cryptosystem: Authentication & Confidentiality



It is also used to exchange a session key, which is a secret key for symmetric encryption generated for use in a particular transaction or session.

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Asymmetric Cryptography

Applications

Hybrid encryption

Due to the high computational cost of public-key cryptography, it's rarely used to encrypt actual data. Instead, a hybrid technique is often used.





Phase 1

Asymmetric Key exchange (e.g., DH)

Public channel

x, g, p		y
$A = g^x \bmod p$	⟷	$B = g^y \bmod p$
$K = B^x \bmod p$		$K = A^y \bmod p$

Phase 2

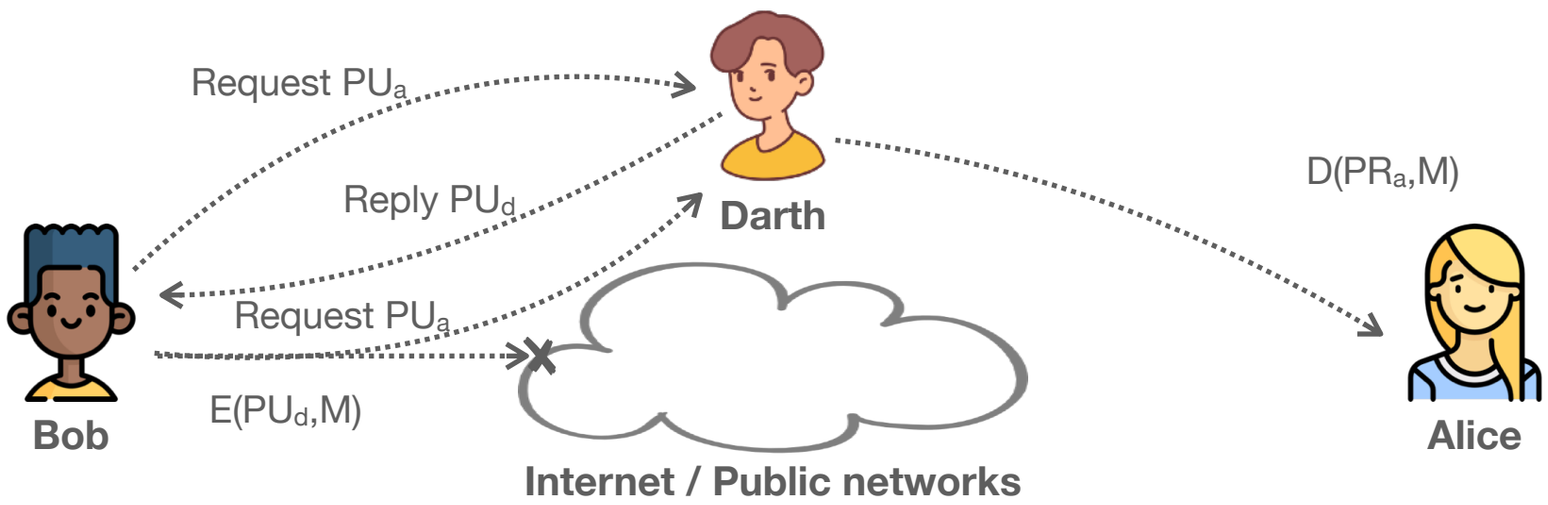
Symmetric encryption (e.g., AES)

$C = E_k(M)$	⟶	$M = D_k(C)$

Asymmetric Cryptography

Man-in-the-Middle (*MitM*) attack

Bob wants to send Alice an encrypted message.



1. Bob sends a request for Alice's public key
2. Darth intercepts the request and replies with his own public key
3. Darth intercepts the message encrypted with PU_d and reads it using PR_d
4. Then, encrypts the message with PU_a and sends it to Alice
5. Alice decrypts the message with PR_a without the knowledge that Darth also had access to the message

It requires a trust mechanism for the association between key pairs and communicating parties.

Asymmetric Cryptography

The RSA algorithm

- Developed in 1977 by Ron **R**ivest, Adir **S**hamir, and Len **A**dleman
- Its security is based on the complexity of factoring and calculating modular logarithms of large numbers.
- It is a block cipher in which the plaintext and ciphertext are integers between `0` and `n-1` for some `n`.
- The general encryption of a message `M` and decryption of a ciphertext `C` follows:

$$C = M^e \bmod n \qquad \textit{e.g.,}\ \texttt{n = 1024}\ \text{bits}$$

$$M = C^d \bmod n = (M^e)^d \bmod n$$

- Both sender and receiver must know the values `n` and `e`, and only the receiver knows the value of `d`.

Public key $PU=\{e,n\}$

Private key $PR=\{d,n\}$

Asymmetric Cryptography

The RSA algorithm

Algorithm requirements

- It is possible to find values of e, d, n such that $M^{ed} \bmod n = M$ for all $M < n$
- It is relatively easy to calculate M^e and C^d for all values of $M < n$
- It is infeasible to determine d given e and n

$$C = M^e \bmod n$$

$$M = C^d \bmod n = (M^e)^d \bmod n$$

Asymmetric Cryptography

The RSA algorithm

Basic number theory

- Two integers a and b are **relatively prime** if and only if their common positive integer factor is 1:

$$gdc(a,b)=1$$ **gdc** - *greatest common divisor*

- **Euler's totient function**: $\varphi(n)$ is the number of positive integers less than n and relatively prime to n. *Ex.* $\varphi(7)=6$ $\{1,2,3,4,5,6\}$
  - For a prime number p, $\varphi(p)=(p-1)$
  - For p, q prime, $\varphi(pq)=(p-1)(q-1)$
- **Fermat's theorem**: if p is prime and a is a positive integer not divisible by p, then

$$a^{p-1} \mod p \equiv 1$$

Generalising Euler, gcd(a,n) = 1

$$a^{\varphi(n)} \mod n \equiv 1$$

Asymmetric Cryptography

The RSA algorithm

Algorithm

Key generation

1. Select p, q (both prime and p≠q)
2. Calculate n=p.q
3. Calculate $\varphi(n)=(p-1)(q-1)$
4. Select integer e ($gcd(\varphi(n),e)=1;\ 1<e<\varphi(n)$)
5. Calculate d ($de \bmod \varphi(n)=1$)

Public key $PU=\{e,n\}$

Private key $PR=\{d,n\}$

Limitation?

Encryption

Plaintext: M<n

Ciphertext: $C = M^e \bmod n$

Decryption

Ciphertext: C

Plaintext: $M = C^d \bmod n$

Asymmetric Cryptography

The RSA algorithm

Algorithm - Example

1. Select p, q (both prime and p≠q)

p = 5, q = 11

2. Calculate n=p.q

n = (5x11) = 55

3. Calculate $\varphi(n)=(p-1)(q-1)$

$\varphi(55) = (p-1)(q-1) = 40$

4. Select integer e ($gcd(\varphi(n),e)=1$; $1<e<\varphi(n)$)

e = 7 ⟶ Public key **PU={7,55}**

5. Calculate d ($d.e \bmod \varphi(n)=1$)

$7.d \bmod 40 = 1$

d=23 ⟶ Private key **PR={23,55}**

Encryption

Plaintext: M = **C** = 2

Ciphertext: **$C = M^e \bmod n$**

$C = 2^7 \bmod 55 = 18 =$ **S**

Decryption

Ciphertext: C = **S** = 18

Plaintext: **$M = C^d \bmod n$**

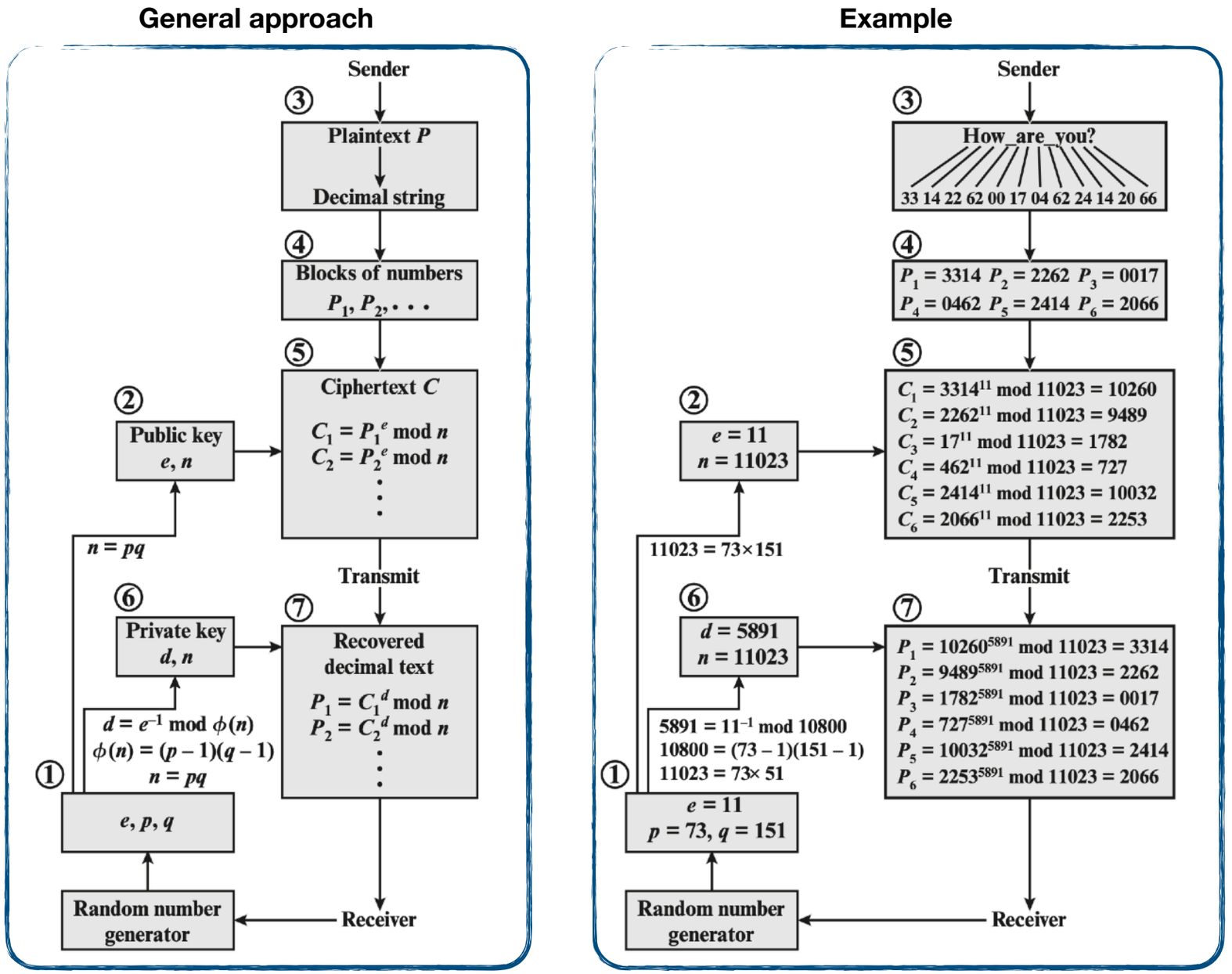
$M = 18^{23} \bmod 55 = 2 =$ **C**

A given message encrypted repeatedly always produces the same ciphertext.

Asymmetric Cryptography

The RSA algorithm

Processing multiple blocks



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